

4.7.5 Atoms into ions and ions into atoms

Metals can be arranged in an activity series in terms of the ability of their atoms to turn into positive ions. Electrolysis is a process that reverses such changes by turning ions back into atoms (and

sometimes molecules). Gas tests are introduced to identify the products of electrolysis. Gain or loss of electrons can be used to classify electrode reactions as reduction or oxidation processes.

The required practical is an investigation of the electrolysis of aqueous solutions using inert electrodes.

4.7.5.1 A reactivity series for metals

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the reactivity of metals with water or dilute acids is related to the tendency of the metal to form its positive ion.	When metals react with other substances the metal atoms form positive ions. Metals can be arranged in order of their reactivity in a reactivity series. The metals potassium, sodium, lithium, calcium, magnesium, zinc, iron and copper can be put in order of their reactivity from their reactions with water and dilute acids. The non-metals hydrogen and carbon are often included in the reactivity series. A more reactive metal can displace a less reactive metal from a compound. The reactions of metals with water and acids are limited to room temperature and do not include reactions with steam.	WS 3.8 Recall and describe the reactions, if any, of potassium, sodium, lithium, calcium, magnesium, zinc, iron and copper with water or dilute acids. WS 3.5 Deduce an order of reactivity of metals based on experimental results. WS 1.2 (HT only) Write ionic equations for displacement reactions.

4.7.5.2 Electrolysis

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe electrolysis in terms of the ions present and reactions at the electrodes. Recall that metals (or hydrogen) are formed at the cathode and non-metals are formed at the anode in electrolysis using inert electrodes.	When an ionic compound is melted or dissolved in water, the ions are free to move about within the liquid or solution. These liquids and solutions are able to conduct electricity and are called electrolytes. Passing an electric current through electrolytes causes the ions to move to the electrodes. Positively charged ions move to the negative electrode (the cathode), and negatively charged ions move to the positive electrode (the anode). Ions are discharged at the electrodes producing elements. This process is called electrolysis. When a simple ionic compound is electrolysed in the molten state using inert electrodes, the metal is produced at the cathode and the non-metal is produced at the anode.	WS 1.2 Predict the products of electrolysis of binary ionic compounds in the molten state. (HT only) Write half equations for the reactions occurring at the electrodes during electrolysis. Students may be required to complete and balance supplied half equations.

4.7.5.3 Electrolysis of aqueous solutions

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe competing reactions in the electrolysis of aqueous solutions of ionic compounds in terms of the different species present.	The ions discharged when an aqueous solution is electrolysed using inert electrodes depend on the relative reactivity of the elements involved. At the negative electrode (cathode), hydrogen is produced if the metal is more reactive than hydrogen. At the positive electrode (anode), oxygen is produced unless the solution contains halide ions when the halogen is produced. This happens because in the aqueous solution water molecules break down producing hydrogen ions and hydroxide ions that are discharged.	WS 1.2 (HT only) Write half equations for the reactions occurring at the electrodes during electrolysis. Students may be required to complete and balance supplied half equations.

Required practical activity 21: investigate what happens when aqueous solutions are electrolysed using inert electrodes. This should be an investigation involving developing a hypothesis.

AT skills covered by this practical activity: chemistry AT 3 and 7.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 177).

4.7.5.4 Tests for gases

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe tests to identify selected gases including oxygen, hydrogen and chlorine.	The test for hydrogen uses a burning splint held at the open end of a test tube of the gas. The test for oxygen uses a glowing splint inserted into a test tube of the gas. The test for chlorine uses damp litmus paper put into chlorine gas.	WS 3.5 Interpret the observations from gas tests.

4.7.5.5 Electron transfer reactions (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain reduction and oxidation in terms of gain or loss of electrons, identifying which species are oxidised and which are reduced.	Oxidation is the loss of electrons and reduction is the gain of electrons. During electrolysis, at the cathode (negative electrode), positively charged ions gain electrons and so the reactions are reductions. At the anode (positive electrode), negatively charged ions lose electrons and so the reactions are oxidations.	WS 4.1 Identify, in a given reaction, symbol equation or half equation, which species are oxidised and which are reduced.